Spam detection

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Overview of problem

- ## Story:

You've just been recruited by the Digital Communications Intelligence Unit (DCIU), a shadow division tasked with protecting humanity from cyber threats. The world is drowning in spam from fake lottery wins to phishing scams. Your mission is clear: build an intelligent system using Python to detect and flag spam messages before they reach unsuspecting inboxes. This is your first real field assignment as a junior data agent.

- ## Business question and tasks

1. What words appear most often in spam vs ham?

2. How accurate is your classifier?

3. Which messages are falsely labeled?

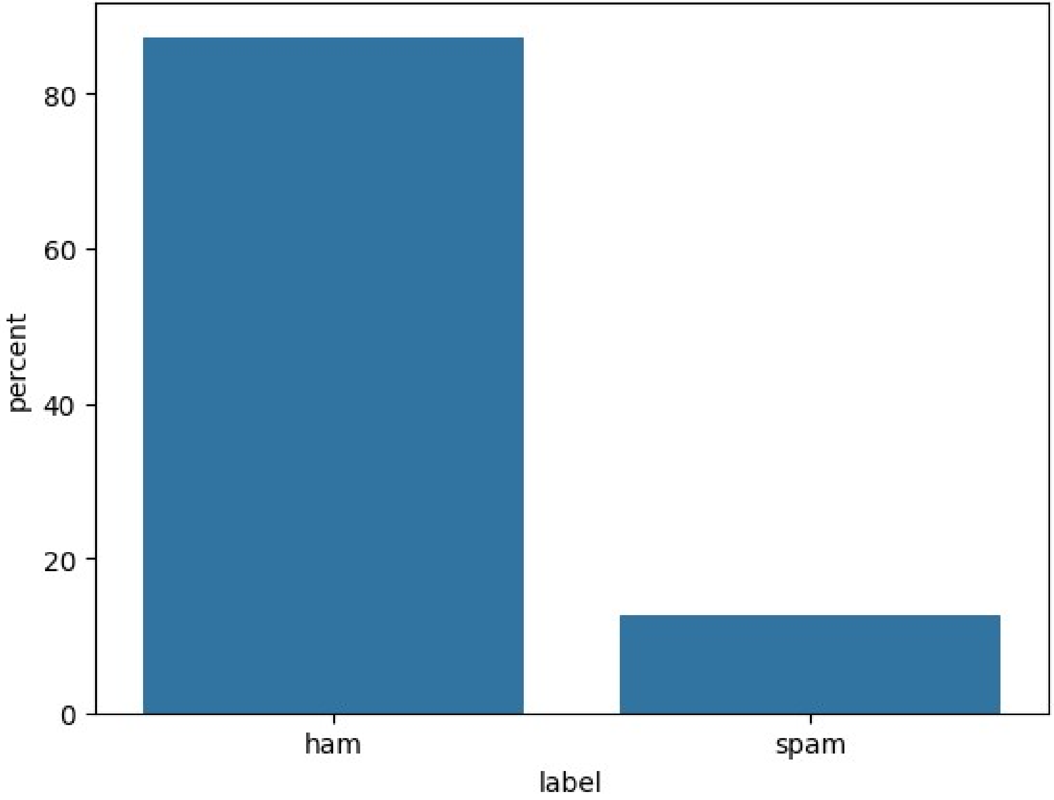
4. Does message length affect likelihood of spam?

5. What are the most predictive keywords or features?

- ## Data: "spam_dataset.csv"

Simple analysis of the data

- **Distribution of labels**
  - Ham messages dominate the dataset (>80%)
  - This is an imbalanced dataset. This property should be taken into account in modeling.



Simple analysis of the data

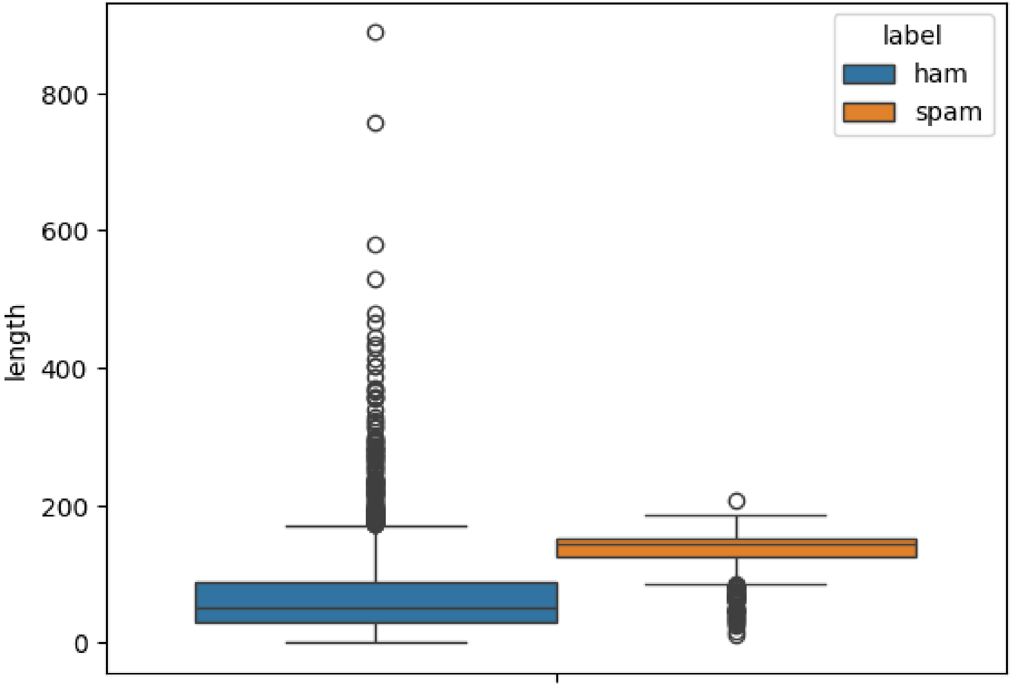
- **Word clouds**

- Some words appearing more frequently in “Spam”: ***free, call, text, txt, reply, mobile, offer***.
- Some words appearing more frequently in “Ham”: ***go, im, ok, come, got, time***.

Simple analysis of the data

- **Length vs. Spam**
  - Message's length has a strong correlation with message's label: Spam messages tend to be significantly longer than ham messages

```
mean   median
label
ham      67.153233    50.0
spam    132.482389   143.0
```



Modeling

- **Two classifiers are built and evaluated:**
  - Naive Bayes (MultinomialNB)
  - Logistic Regression

- **Classifiers are optimized using grid search. The best models are evaluated using f1 score**

Modeling

- ## Classification reports:

Naive Bayes:

	precision	recall	f1-score	support
ham	0.978	0.998	0.988	903
spam	0.982	0.847	0.910	131
accuracy			0.979	1034
macro avg	0.980	0.923	0.949	1034
weighted avg	0.979	0.979	0.978	1034

Logistic regression:

	precision	recall	f1-score	support
ham	0.984	0.996	0.990	903
spam	0.967	0.885	0.924	131
accuracy			0.982	1034
macro avg	0.975	0.941	0.957	1034
weighted avg	0.981	0.982	0.981	1034

Observations:

1. Naive Bayes and Logistic Regression have the relatively similar performance on ham messages.

2. On spam messages:

- ***Precision:*** Naive Bayes acts slightly better than Logistic Regression. For Naive Bayes, 2.8% of spam-labeled messages are actually hams. For Logistic Regression, 3.3% of spam-labeled messages are actually hams.

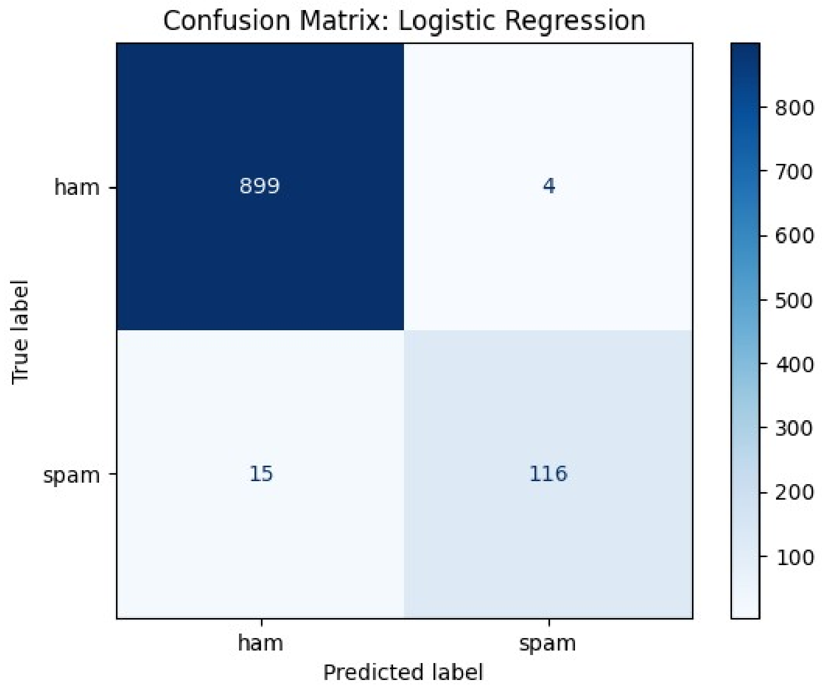
- ***Recall***: On this measure, Logistic Regression acts better than Naive Bayes. LR successfully catches about 88.5% of all the spam messages, while NB is less effective (84.7%).

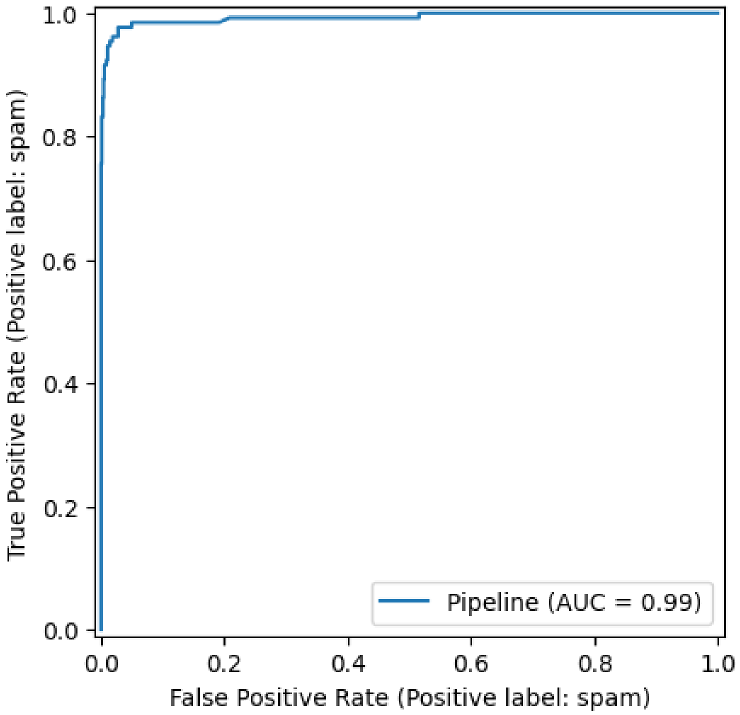
- ***F1-score:*** Logistic Regression's score of 0.924 proves it handles the trade-off between "missing spam" and "false alarms" more effectively than Naive Bayes (0.910).

Conclusion: In overall, Logistic Regression is better for production than Naive Bayes for spam detection.

Modeling

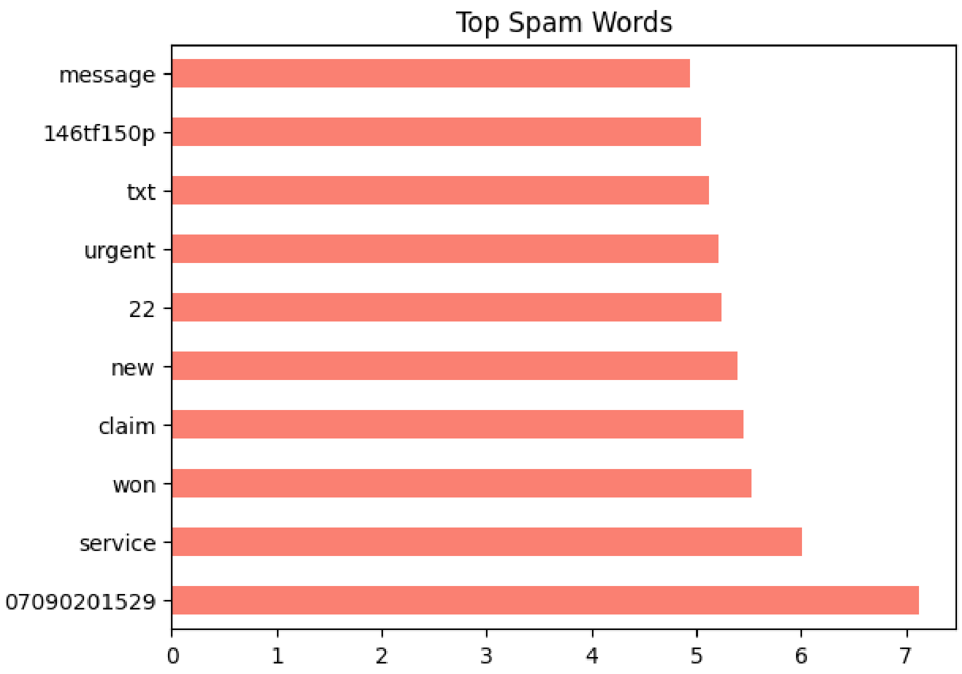
- **More on Logistic Regression: nice model**

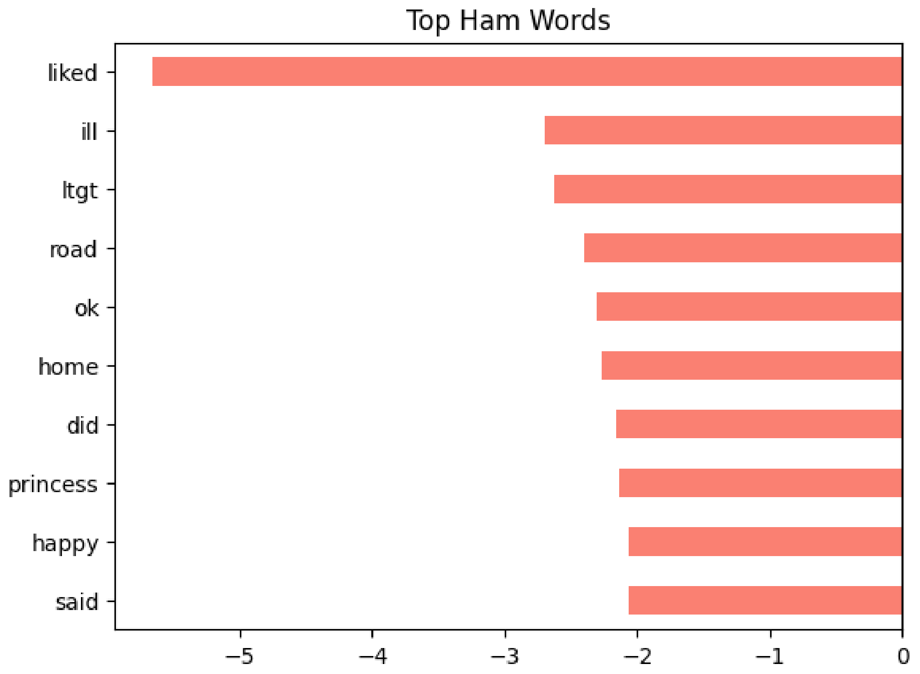




Modeling

- **Top predictive words for Spam and Ham from the best logistic classifier**





What was learned?

- Text preprocessing
- Text tokenization and vectorization
- Classification modeling (Logistic Regression and Naive Bayes)
- Grid search for model optimization
- Classification metrics: precision, recall, f1